
The Two-Bin Revolution

Optimizing solid
waste segregation

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Abstract

In this project, we have addressed one of the prevalent issues across the globe-Inadequate waste management and segregation systems.

Waste Management and Segregation refers to the process of collecting, treating, and disposing of various types of waste materials in an environmentally responsible manner.

It plays a crucial role in our society for numerous reasons. It not only ensures the proper disposal and recycling of waste but also helps protect the environment, conserve resources, and promote sustainable development. By managing waste effectively, we can reduce pollution levels, mitigate health risks, and create a cleaner and healthier living environment for future generations

We embarked on this project with the aim of designing a sustainable waste segregation tool that is cost effective, feasible and can be put into practice



Problem Statement

Inefficient Waste Segregation Methods in Urban Regions

India's material consumption has increased six-fold between 1970 and 2015 and is expected to double again by 2030. With rapid urban growth and lifestyle changes, the country is facing a mounting waste crisis. Managing this increasing volume of waste poses operational and financial challenges for urban and rural areas.

As a mitigatory action, source segregation and scientific disposal of DRY WASTE, which makes up about 35% of the total waste stream, with plastic being a major concern in India is essential

Motivation and Background

Mixing wet and dry waste has severe environmental, economic, and health-related consequences. When these two types of waste are combined, they contaminate each other, making resource recovery nearly impossible and leading to widespread pollution.

Here is a detailed breakdown of impacts:

1. Environmental Impacts

- Destroys Recycling: Wet waste contaminates dry recyclables, sending valuable materials to landfills.
- Generates Potent Greenhouse Gases: Anaerobic decomposition of organic waste in landfills releases methane, a powerful greenhouse gas.
- Creates Toxic "Garbage Juice" (Leachate): Moisture from wet waste mixes with other toxins, forming leachate that contaminates groundwater and soil.
- Increases Air Pollution: Burning mixed waste in dumpsites releases toxic fumes, including carcinogens.

2. Economic Impacts

- Loses Valuable Resources: Unsegregated waste loses its potential for composting, biogas, or recycling, becoming a costly liability.
- Increases Operational Costs: Transporting and managing mixed waste is expensive and inefficient for municipalities, increasing landfill fees.
- High Remediation Costs: Cleaning up pollution from contaminated sites is a massive, taxpayer-funded expense.

3. Health and Social Impacts

- Breeding Ground for Disease: Mixed waste attracts pests, spreading diseases like cholera, typhoid, and dengue.
- Direct Health Hazards: People near landfills suffer from air and water

pollution, leading to respiratory and skin issues.

- Risks to Waste Workers: Sanitation workers face high risks of injury and exposure to hazardous materials and pathogens.

In summary, mixing wet and dry waste turns valuable resources into a toxic and expensive environmental problem. Segregation at the source is the single most effective step in solving this issue.

Introduction

Solid and wet waste segregation is the process of separating waste into two main streams: wet waste, which is organic and biodegradable (like food scraps and yard trimmings), and dry waste, which includes non-biodegradable materials (like paper, plastic, and metal). This separation is crucial because wet waste can be COMPOSTED , while dry waste can be RECYCLED , which reduces landfill waste and pollution, conserves resources, and improves sanitation.

At present, waste segregation is done by the following methods-

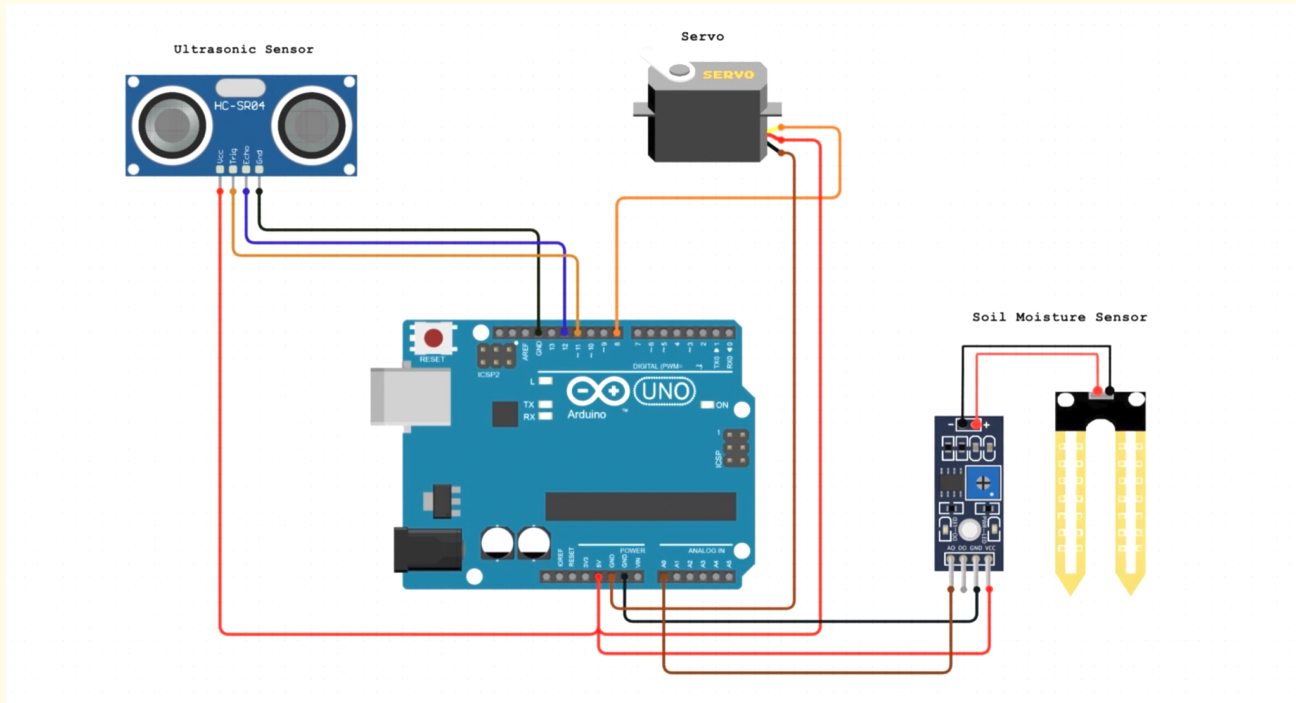
Manual Sorting, where waste pickers and workers manually sort mixed waste at collection centers.

Manual hand-picking , where sorters on a conveyor belt remove visually identifiable items

Automated Sorting , which are advanced machineries used to sort waste . This includes:

- Trommel screens
- Sensors
- Use of AI

Circuit Diagram



Project Description

This project is a practical example of how electronics can be used to create an automated solution for efficient waste management at home. The components used are listed below.

Electronic Components (The "Smart" System)

- **Arduino Uno:** The blue microcontroller board that acts as the central brain, controlling the sensors and motors.
- **HC-SR04 Ultrasonic Sensor:** This sensor (with two "eyes") is placed on the front. It detects when a hand approaches the bin, likely to activate the system.
- **Soil Moisture Sensor:** This is the key sensor for segregation. It's the component with two prongs on which the waste is first placed. It detects the moisture level to determine if the waste is "wet" or "dry."
- **IR (Infrared) Sensor:** A small sensor used to detect the presence of an object (the waste) on the sorting flap before the mechanism activates.
- **Servo Motors (x3):** Small blue motors (likely SG90 type) are used for all the moving parts.
 - One servo controls the sorting flap, tilting it to drop waste into the correct bin.
 - Two servos are used to automatically lift and close the lids for the "WET" and "DRY" compartments.
- **Power Source:** A **9V battery** with a battery snap connector is used to power the Arduino and the motors.
- **Jumper Wires:** A breadboard and various male-to-female and male-to-male jumper wires are used to connect all the electronic components.

SDG Goals addressed in the project



Waste segregation is included in the Sustainable Development Goals (SDGs) through its direct connection to multiple goals, most notably

- SDG 11 (Sustainable Cities and Communities)
- SDG 12 (Responsible Consumption and Production)

Effective waste management, which includes segregation, is essential for improving urban living, reducing pollution, and promoting resource efficiency and circular economy models.

Additional Insights

SDG 11: Sustainable Cities and communities

Proper waste segregation improves the cleanliness, health and livability of urban areas

SDG 12: Responsible Consumption and production

Segregation is key to enabling recycling, reuse and resource recovery , which reduces the consumption of raw materials and minimizes waste generation

SDG 14: Life Below Water

Proper segregation prevents waste, especially plastics, from polluting oceans and harming marine life

“Educate , Segregate , Recycle- the golden triangle of waste management”

Conclusion and future scope

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Waste is a design flaw, the ability to create circularity of all materials eliminates waste and builds a sustainable world.

~ Kate Krebs



“This project highlights the importance of introducing two bin systems and introducing it in urban centres to optimise waste segregation”

The future scope for a smart dustbin. The potential developments can be grouped into advanced in-bin features, network-level integrations, and new applications.

1. Advanced In-Bin Features

The next generation of smart bins will move from simple wet/dry sorting to more complex, automated tasks.

- **AI-Powered Waste Recognition:** Instead of just detecting moisture, future bins will use built-in cameras and AI (Artificial Intelligence) with computer vision. The AI would be trained to identify *exactly* what an item is (e.g., "PET plastic bottle," "aluminum can," "apple core") with high accuracy.
- **Multi-Category Sorting:** With AI recognition, the bin can automatically sort waste into multiple internal compartments, not just two. For example, it could have separate bins for:
 - Organics (Wet Waste)
 - Paper/Cardboard
 - Plastics
 - Metals (Aluminum/Steel)
 - Glass
 - E-waste or batteries
- **Automatic Compression:** To reduce the frequency of emptying, integrated compactors can automatically crush dry waste like plastics and paper, increasing the bin's capacity.

- **Odor Control & Hygiene:** Features like odor-neutralizing filters and automated self-cleaning mechanisms will become more common, especially in residential and commercial settings.
- **Onboard Power:** Many public smart bins are already being developed with solar panels on top to power the sensors, compactors, and communication hardware, making them self-sufficient and sustainable.

2. IoT and Smart City Integration

The true "future scope" is connecting the bin to a larger network (the Internet of Things, or IoT) to optimize waste management on a city-wide scale.

- **Real-Time Fill-Level Monitoring:** Bins will use ultrasonic sensors to monitor how full they are in real-time.
- **Automated Collection Alerts:** When a bin is full, it will automatically send a notification to the municipal waste collection service.
- **Dynamic Route Optimization:** This is the key application. Instead of collecting every bin on a fixed schedule, AI algorithms will use real-time data from all smart bins to calculate the most efficient collection route for the garbage trucks, collecting *only* the bins that are full. This saves immense amounts of fuel, labor, and time.
- **Data Analytics for Urban Planning:** The data collected from smart bins provides powerful insights. City planners can analyze this data to:
 - Identify high-waste-generation areas.

- Understand waste generation patterns (e.g., more plastic waste near parks).
- Make informed decisions about where to place new bins or recycling centers.
- **Blockchain for Transparency:** For managing hazardous or e-waste, blockchain technology can be used to create a transparent, tamper-proof record of the waste from collection to final disposal, preventing illegal dumping.

3. New Applications and User Interaction

The technology can be adapted for different environments and business models.

- **Gamification and Incentives:** In public spaces, smart bins could feature interactive screens. These screens could be used for advertising or for reward systems (e.g., a "reverse vending machine") where users get a small reward (like a coupon) for correctly recycling an item.
- **Autonomous Collection:** The "bin" itself could become mobile. Research is underway on autonomous garbage collection robots that can navigate urban areas, collect waste, and even perform real-time sorting using robotic arms and computer vision.
- **Specialized Environments:** This technology is highly valuable in places with specific waste needs, such as:
 - **Hospitals:** For safely sorting and tracking different types of medical and hazardous waste.
 - **Commercial Kitchens:** For tracking food waste to optimize purchasing and reduce costs.